

Prathamesh Bhavsar

Dallas, TX (Open to Relocation) | prathameshbhavsar19@gmail.com | +1(940)279-9009 | GitHub | Website | LinkedIn

Summary

AI Engineer with 4 years of experience building GenAI-powered tools and LLM agents for enterprise software delivery teams. Skilled in RAG architectures, prompt engineering, and agentic workflows using GPT-4, LangChain, and open-source foundation models, with hands-on experience in governance-aware and compliance-driven AI deployments at scale.

Skills

- **Programming & Databases** - Python, SQL, Spark, Snowflake, MongoDB, TensorFlow, PyTorch, NumPy, Pandas
- **AI & ML Skills** - LLMs, LangChain, RAG, Prompt Engineering, Deep Learning, NLP, Computer Vision
- **Soft Skills** - Cross-functional Collaboration, Problem Solving, Analytical Thinking, Communication

Experience

Intern, Tenet Healthcare - Dallas, TX Aug 2025 – Present

- Architected an ICD-10 medical coding agent using LangChain and GPT-4 with **FAISS RAG** retrieval on GCP Cloud Run with BigQuery logging and A/B testing, achieving **89%** accuracy and cutting manual coding time by **55%** across **12 facilities**.
- Developed a clinical trial matching agent using BioBERT and LangChain with Neo4j graph traversal across **500+ trial criteria**, deployed on Cloud Run with HIPAA-compliant handling that boosted enrollment by **42%** and compressed screening from 3 hours to **15 minutes**.
- Engineered a real-time clinical decision support system using TensorFlow and FHIR API streams with Redis caching on Kubernetes and Prometheus monitoring, detecting critical events across **8 ICUs** and lowering mortality by **12%**.

Software Development Engineer (Machine Learning), EnelX - Mumbai, India Jul 2021 – Aug 2024

- Built a predictive maintenance system using TensorFlow LSTM and Kafka with a LangChain LLM agent on SageMaker and Kubernetes with MLflow tracking, slashing outages by **40%** and trimming maintenance costs by **25%**.
- Designed an AI-powered energy audit engine using GPT-4 and LangChain on AWS with ECS, RDS, and S3, automating **80%** of audit workflows and uncovering **24%** average savings across **300+ facilities**.
- Constructed an outage root cause analysis system pairing LSTM detection on SCADA streams via Kafka with Neo4j knowledge graph and LangChain reasoning with Grafana dashboards, compressing identification from 4.2 hours to **35 minutes** across **2,000+ grid assets**.
- Created an EV fleet charging optimizer using XGBoost and PuLP on AWS, ingested via PySpark on EMR and Redshift, trained on SageMaker with MLflow, deployed on ECS via FastAPI that cut costs by **18%** and lifted uptime to **97%** across **800+ vehicles**.
- Spearheaded billing analysis across **30+ suppliers** using Amazon Textract and SageMaker MLOps workflows, reducing duplicate overhead from 12% to **5%** and saving **50+ hours monthly**.

Education

University of Texas at Dallas, MS in Computer Engineering Aug 2024 – May 2026

University of Mumbai, Bachelor of Engineering in Information Technology Jul 2017 – Jun 2021

Projects

AgentToyota — Smart Vehicle Financing AI Agent (FastAPI, React, Langchain) ([GitHub Link](#))

- Built an app for personalized **financing/leasing** using **FastAPI+React** with **EMI simulations**, comparisons and model picks. **UUID** login on **SQLite**. **LLM** explanations cut decision time **45%** and raised completion **22%**.

Doclytics (RAG, FAISS, Azure OpenAI, Streamlit) ([GitHub Link](#))

- Created a **Streamlit**-based **RAG** pipeline to parse LTIMindtree reports, extract structured data from PDFs via **PyMuPDF** and **tabula-py**, embed with **SentenceTransformers**, retrieve with **FAISS**, and query **GPT-4o** for cited, chart-enhanced responses.

GenAI LinkedIn Post Generator (Llama, Langchain, Streamlit) ([GitHub Link](#))

- Designed a LinkedIn content generator using **LLama** on Groq API and **Langchain**; extracted metadata like tone and topic to generate personalized posts via few-shot learning, deployed with **Streamlit** UI, and accelerated content creation by **60%**.

StreamFraud: Real-Time Credit Card Fraud Detection (Kafka, Docker, Isolation Forest) ([GitHub Link](#))

- Built a real-time credit card fraud detection pipeline using **Apache Kafka** (with Zookeeper), trained an **Isolation Forest** model on the dataset, and deployed a live **Streamlit** dashboard; fully containerized with **Docker Compose**, achieving **99.79%** accuracy.